B-trees exercises

18-E01 Node capacity

Difficulty: Basic

Minimum degree t=4. Find nonroot key bounds and maximum child pointers. Is a leaf with6 keys legal?

18-E02 Root split

Difficulty: Advanced

A t=2 full root holds[10,20,30]. Split before inserting25. Give the new root/children and explain which side receives25.

18-E03 I/O height

Difficulty: Research

For t=100,n=10^9 keys,use height and fanout constraints to estimate nodes read during search.

18-E04 Borrowing must retain subtrees

Difficulty: Advanced

At t=2,root[20] has left internal[10] with leaves[5],[15],and right internal[30,40] with leaves[25],[35],[45]. Delete5,listing borrowing before descent,next-level merge,and final tree. Where must the[25] pointer move?

18-E05 Absent deletion may change shape

Difficulty: Research

At t=2,root[20] has leaves[5],[30]. Delete absent7 using repair-before-descent. Draw the result and prove contraction remains legal despite NOT_FOUND. What must stay unchanged?

18-E06 Separate page comparisons from I/O

Difficulty: Research

A B-tree has n keys,height h,minimum degree t,one node per page,resident root,and initially uncached other pages. Compare linear and binary within-page search for page reads and CPU comparisons. Why does binary search not automatically make insertion CPU O(h log t)?

B-trees exercises (continued)

> Complete your derivations before opening the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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